

Practice Quiz: Action (Computational Neuroscience)

Name: __ Date: __

Part A: Multiple Choice

1. In Active Inference, the motor cortex sends: A) Direct commands to muscles B) Proprioceptive predictions that spinal reflexes act to fulfill C) Random signals that muscles interpret D) Only inhibitory signals
2. The basal ganglia direct pathway: A) Suppresses all movement B) Facilitates selected actions by releasing them through the thalamus C) Only processes sensory information D) Is responsible for memory
3. Dopamine's role in the basal ganglia is to: A) Strengthen the No-Go pathway B) Tip the balance toward the Go pathway, favoring action initiation C) Block all neural activity D) Only affect emotional processing
4. Climbing fibers in the cerebellum carry: A) Motor commands to muscles B) Error signals from the inferior olive that correct forward model predictions C) Visual information D) Pleasure signals
5. Mirror neurons fire when: A) You look in a mirror B) Both when you perform an action and when you observe someone else performing the same action C) Only during sleep D) Only during intense concentration
6. Huntington's disease causes involuntary movements because: A) The motor cortex is damaged B) Loss of indirect (No-Go) pathway neurons means competing actions are not suppressed C) The cerebellum is too active D) Too much dopamine is produced
7. Motor learning in the cerebellum involves: A) Growing new neurons B) Modifying synapses between parallel fibers and Purkinje cells based on climbing fiber error signals C) Memorizing conscious rules about movement D) Increasing blood flow

Part B: Short Answer

1. Explain why a patient with Parkinson's disease can sometimes move normally when catching a ball thrown unexpectedly (the "kinesia paradoxa" phenomenon). Use the distinction between basal ganglia policy selection and direct reflex-based movement.
2. Trace the cerebellar learning circuit for learning to ride a bicycle. What are the initial prediction errors? How do they change over weeks of practice? What role do climbing fibers play?
3. How do mirror neurons support Active Inference's claim that perception and action are fundamentally the same process? Explain using the concept of generative models.